MUSIIWA THANDABANTU
musiiwathandy@gmail.com
0763278611

DATA ANALYTICS

Database : employees_db

id	first_name	last_name	department	Salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San francisco

Count() function

① Write a SQL query to find the total number of employees in the company

```
SELECT COUNT(id) AS total_number_of_employees
FROM 'work_employee'.'default'.'employees';
```

total_number_of_employees
10

② Sum() function

Write a SQL query to find the total salary paid to all employees in the IT department

Sum() function

② Write a SQL query to find the total salary paid to all employees in the IT department

```
SELECT department, Sum(salary) AS 'total_salary'
FROM 'work-employee'.'default'.'employees'
WHERE department = 'IT'
GROUP BY department;
```

department	total salary
IT	220000

AVG() function

③ Write a SQL query to calculate the average salary of employees in the HR department

```
SELECT department, AVG(salary) AS 'average_salary'
FROM 'work_employee'.'default'.'employees'
WHERE department = 'HR'
GROUP BY department;
```

department	average_salary
HR	49500

MIN() and MAX() function

④ Write a SQL query to find the highest and lowest salary in the company

```
SELECT MIN(salary), MAX(salary)
FROM 'work_employee'.'default'.'employees';
```

min(salary)	max(salary)
48000	62000

GROUP BY Statement

⑤ Write a SQL query to group employees by department and display the total salary paid in each department

```
SELECT department, Sum(salary) AS total_salary
FROM 'work_employee'.'default'.'employees'
GROUP BY department;
```

	department	total_salary
1	IT	220000
2	HR	99000
3	Finance	119000
4	Marketing	105000

GROUP BY and COUNT()

⑥ Write a SQL query to count how many employees work in each city

```
SELECT city, COUNT(city) AS employees
FROM 'work_employee'.'default'.'employees'
GROUP BY city;
```

City	employees
New York	2
Chicago	3
Los Angeles	2
San francisco	2
Houston	1

GROUP BY and ORDER BY

⑦ Write a SQL query to group employees by department, calculate the average salary in each department and order the results in descending order of average salary

```
SELECT department, AVG(salary)
FROM 'work-employee'.'default'.'employees'
GROUP BY department
ORDER BY AVG(salary) DESC;
```

department	avg(salary)
Finance	59500
IT	55000
Marketing	52500
HR	49500

HAVING Clause

⑧ Write a SQL query to find department where the total salary paid exceeds 100,000 (use GROUP BY and HAVING)

```
SELECT department, Sum(salary) As 'total salary'
FROM 'work-employee'.'default'.'employees'
GROUP BY department
HAVING Sum(salary) > 100000;
```

department	total salary
IT	220000
Finance	119000
Marketing	105000

Combining GROUP BY, HAVING and ORDER BY

(a) Write a SQL query to list cities where more than one employee works, ordered by the number of employees in descending order

```
SELECT city, COUNT(city) AS employees
FROM 'work-employee'.'default'.'employees'
GROUP BY city
HAVING COUNT(city) > 1
ORDER BY employees DESC;
```

City	employees
Chicago	3
New York	2
Los Angeles	2
San francisco	2

Combining Aggregate functions

(b) Write a SQL query to find the department with the highest average salary

```
SELECT department, AVG(salary) AS 'highest average salary'
FROM 'work-employee'.'default'.'employees'
GROUP BY department;
```

department	highest average salary
IT	55000
HR	49500
Finance	59500
marketing	52500